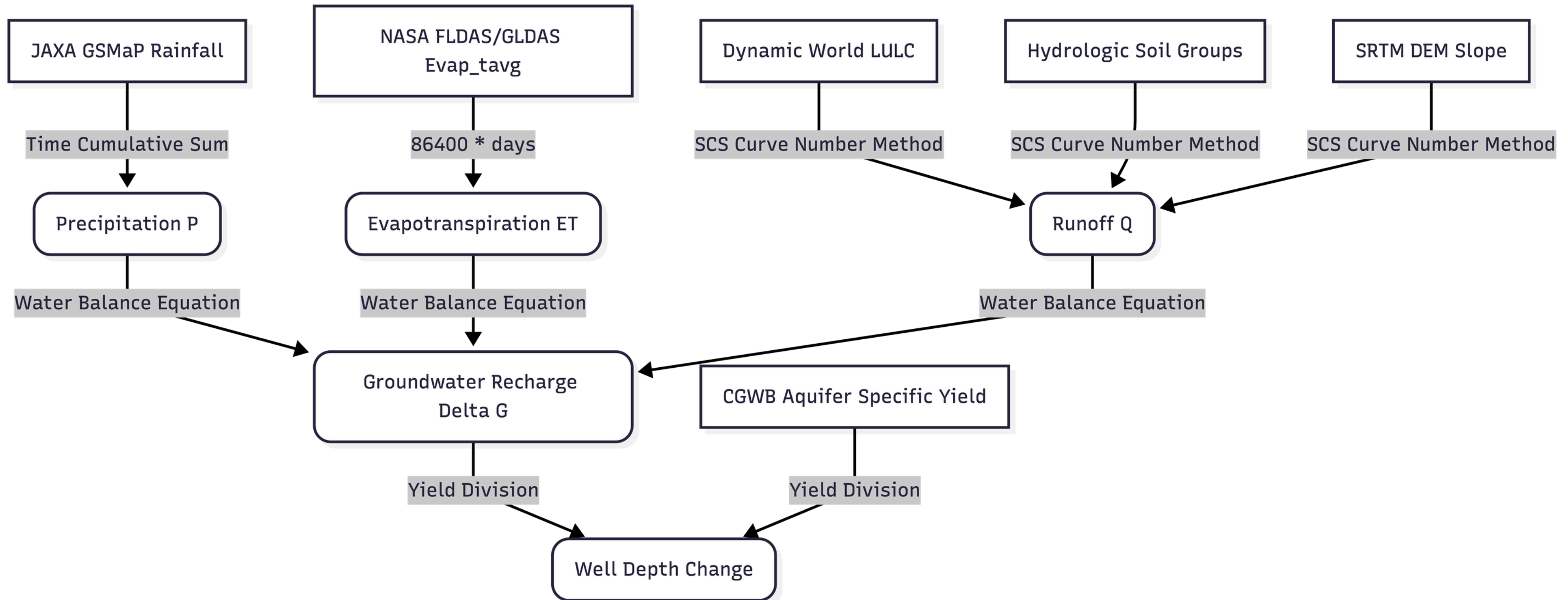


Hydrological & Water Elements

- Precipitation (P): The amount of rainfall entering the watershed (modeled using JAXA GSMaP satellite telemetry).
- Evapotranspiration (ET): The amount of water moving back into the atmosphere via soil evaporation and plant transpiration (modeled using NASA FLDAS/GLDAS).
- Surface Runoff (Q): The volume of excess rainfall flowing over land surfaces (computed via a slope-adjusted SCS-CN model).
- Groundwater Recharge (Delta G): Net water replenishing the subterranean water table.
- Aquifers: Subterranean water-bearing rock layers, categorized by their geographical boundaries and Specific Yield (Sy) values.
- Well Depths: The vertical distance from the ground surface to the local groundwater level.
- Surface Water Bodies (SWB): Spatial outlines of lakes, rivers, reservoirs, and natural ponds.
- Drainage Lines & Streams: Map paths of natural water flow channels.
- Stream Order: Strahler ordering class of stream paths (e.g., first-order tributary streams vs. higher-order main rivers) used to prioritize water harvesting sites.

Current Working



Terrain & Geological Elements

- Digital Elevation Model (DEM): High-resolution representation of topographical relief (SRTM / FABDEM layers).
- Slope: The physical angle/gradient of the land surface (used to determine soil erosion risk and site runoff speed).
- Lithology: The underlying rock type composition (e.g., sandstone, basalt, alluvial clay, granitic gneiss) which dictates the soil's natural permeability and infiltration rates.
- Lineaments: Geological structural faults, joints, or fractures in the Earth's crust that act as pathways for subterranean water movement.

Vegetative & Forest Elements

- Tree Canopy Cover Density (CCD): The percentage of forest/vegetation cover shielding the ground.
- Tree Canopy Height: The vertical height of tree stands, reflecting biomass and forest maturity.
- Normalized Difference Vegetation Index (NDVI): Reflects the density and photosynthetic health of green vegetation.
- Land Use Land Cover (LULC): Classifies the physical cover of the Earth's surface into natural categories:
 - Forest & Trees
 - Grasslands / Pastures
 - Shrubland & Scrub
 - Croplands (distinguishing between Kharif/Rabi cropping intensity)
 - Wetlands & Water
 - Bare Soil / Rock / Sandy plains

Climate & Meteorological Anomalies

- Dry Spells: Runs of consecutive dry days indicating prolonged dry weather during critical crop growth windows.
- Drought severity: Evaluates meteorological drought (extreme lack of rain) and agricultural drought (soil moisture and vegetation stress).

